# ASSIGNMENT 2: PYTHON & BASH

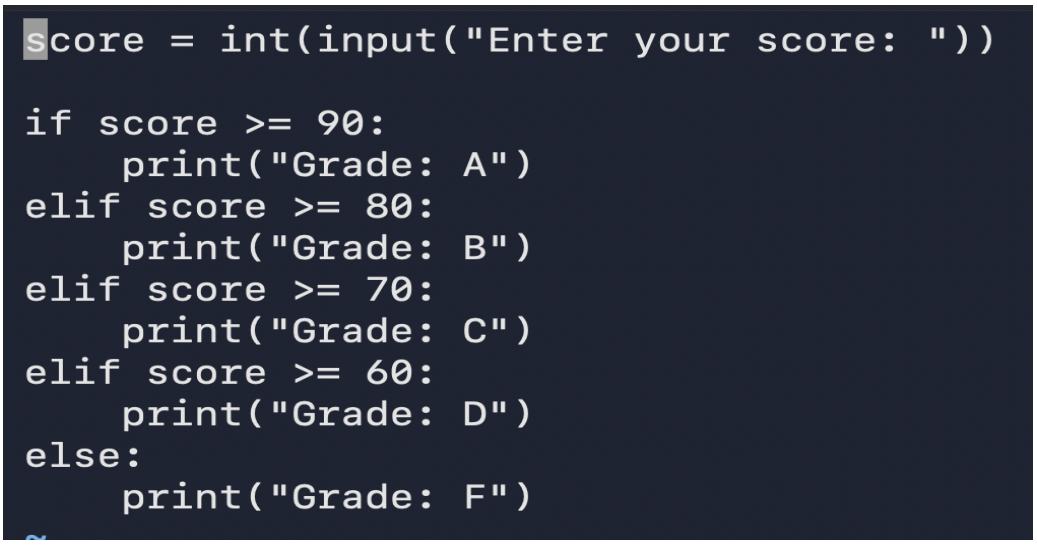
## 1. Grade Checker

Take a score as input and print the grade based on the following:

1. **90+ :** “A”
2. **80–89 :** “B”
3. **70–79 :** “C”
4. **60–69 :** “D”
5. **Below 60 :** “F”
6. Use a basic `if / else` statement to determine the grade.

**Output**:

```
score = int(input("Enter your score: "))
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
elif score >= 60:
    print("Grade: D")
else:
    print("Grade: F")
```



```
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 grade_checker.py
Enter your score: 85
Grade: B
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 grade_checker.py
Enter your score: 92
Grade: A
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 grade_checker.py
Enter your score: 75
Grade: C
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 grade_checker.py
Enter your score: 64
Grade: D
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 grade_checker.py
Enter your score: 58
Grade: F
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 grade_checker.py
Enter your score: 44
Grade: F
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
```

**Explanation:**

- `input()` is used to accept the student's score from the user.
- `int()` converts the entered score into an integer.
- The `if` statement checks whether the score is 90 or above and assigns Grade A.
- The `elif` statements check the remaining score ranges to assign Grades B, C, and D.
- The `else` statement assigns Grade F if the score is below 60.
- `print()` displays the final grade.

**Result:** The program successfully determines and displays the student's grade based on the entered score. For example, if the user enters 95, the program displays Grade A.

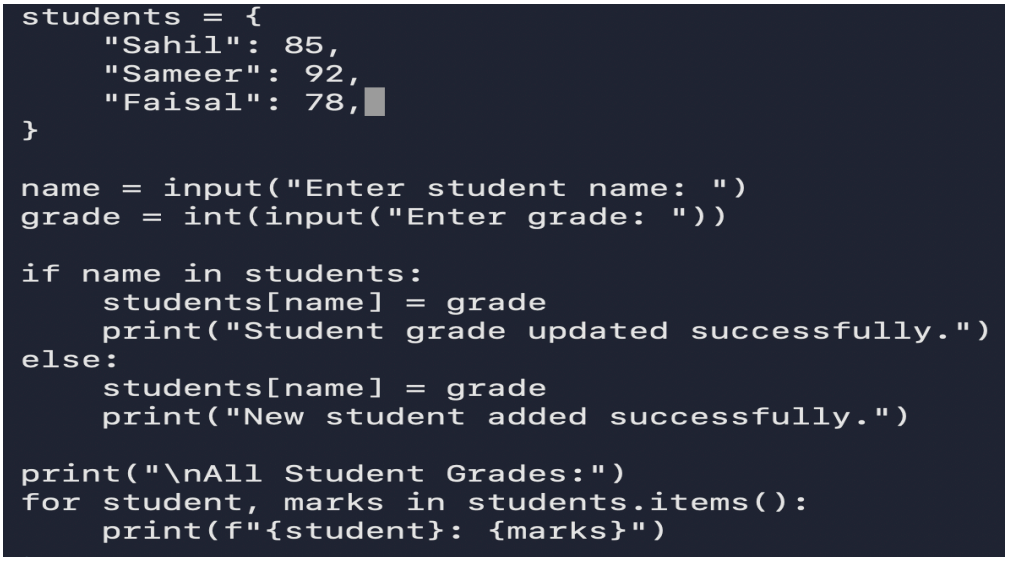
## 2. Student Grades

Create a dictionary where the keys are student names and the values are their grades. Allow the user to:

1. Add a new student and grade.
2. Update an existing student's grade.
3. Print all student grades.
4. Use a `dictionary` and basic operations with if / else.

**Output**:

```
students = {
    "Sahil": 85,
    "Sameer": 92,
    "Faisal": 78,
}
name = input("Enter student name: ")
grade = int(input("Enter grade: "))
if name in students:
    students[name] = grade
    print("Student grade updated successfully.")
else:
    students[name] = grade
    print("New student added successfully.")
print("\nAll Student Grades:")
for student, marks in students.items():
    print(f"{student}: {marks}")
```



```
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 student_grades.py
Enter student name: Rohit
Enter grade: 88
New student added successfully.

All Student Grades:
Sahil: 85
Sameer: 92
Faisal: 78
Rohit: 88
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
```

**Explanation:**

- A dictionary named `students` is created to store student names as keys and their grades as values.
- `input()` accepts a student's name and grade from the user.
- The `if` statement checks whether the student already exists in the dictionary.
- If the student exists, the program updates the student's grade.
- Otherwise, the program adds the new student and grade to the dictionary.
- The `for` loop with `.items()` displays all student names and their grades.

**Result:** The program successfully adds new students, updates existing students' grades, and displays all student records stored in the dictionary.

### 3. Write to a File

1. Write a program to create a text file and write some content to it, using file functions like `open()` and write().

**Output**:

file = open("sample.txt", "w")
file.write("Hello, this is my first file.\n")
file.write("I am learning Python file handling.")
file.close()
print("Data written to file successfully.")

```
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file.write("Hello, this is my first file.\n")
file.write("I am learning Python file handling.")
file.close()

print("Data written to file successfully.")
~
```

```
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 main.py
Data written to file successfully.
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % ls -ltr
total 32
-rw-r--r--  1 rohitbhupat  staff  228 21 Oct 14:51 grade_checker.py
-rw-r--r--  1 rohitbhupat  staff  422 21 Oct 14:54 student_grades.py
-rw-r--r--  1 rohitbhupat  staff  185 21 Oct 14:58 main.py
-rw-r--r--  1 rohitbhupat  staff   65 21 Oct 14:59 sample.txt
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % cat sample.txt
Hello, this is my first file.
I am learning Python file handling.%
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
```

**Explanation:**

- `open("sample.txt", "w")` creates a file named `sample.txt` in write mode. If the file already exists, its previous contents are overwritten.
- The `write()` function writes text into the file.
- The `\n` character moves the second line of text to a new line.
- `close()` closes the file after writing is complete.
- `print()` displays a confirmation message in the terminal.

**Result:** The program successfully creates the text file and writes the specified content into it. The terminal displays the message "Data written to file successfully."

## 4. Read from a File

1. Open the file in read mode and use `file.read()` to read and display its content.

**Output**:
file = open("sample.txt", "r")
content = file.read()
print("File Contents:")
print(content)
file.close()

```
file = open("sample.txt", "r")
content = file.read()
print("File Contents:")
print(content)
file.close()
```

```
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh % python3 read_file.py
File Contents:
Hello, this is my first file.
I am learning Python file handling.
rohitbhupat@Rohits-MacBook-Air Python_Bash_SahilShaikh %
```

**Explanation:**

- `open("sample.txt", "r")` opens the existing file in read mode.
- `read()` reads the entire contents of the file.
- The variable `content` stores the text read from the file.
- `print()` displays the file contents in the terminal.
- `close()` closes the file after reading is complete.

**Result:** The program successfully reads and displays the contents of `sample.txt` in the terminal. It demonstrates how Python can retrieve and display data stored in a text file.